

Datasets and tokenisation

From text to token IDs to training examples

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Section 1

Text becomes numbers

A model reads token IDs

Text Once upon a time, there was a little girl named Lily.

A model reads token IDs

Text Once upon a time, there was a little girl named Lily.

Tokens

Once	␣upon	␣a	␣time	,	␣there	␣was	␣a	␣little	␣girl	␣named	␣Lily	.
------	-------	----	-------	---	--------	------	----	---------	-------	--------	-------	---

A model reads token IDs

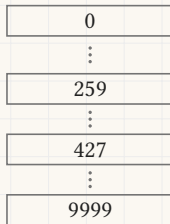
Text Once upon a time, there was a little girl named Lily.

Tokens	Once	␣upon	␣a	␣time	,	␣there	␣was	␣a	␣little	␣girl	␣named	␣Lily	.
IDs	427	437	259	396	44	401	283	259	389	473	402	407	46

A model reads token IDs

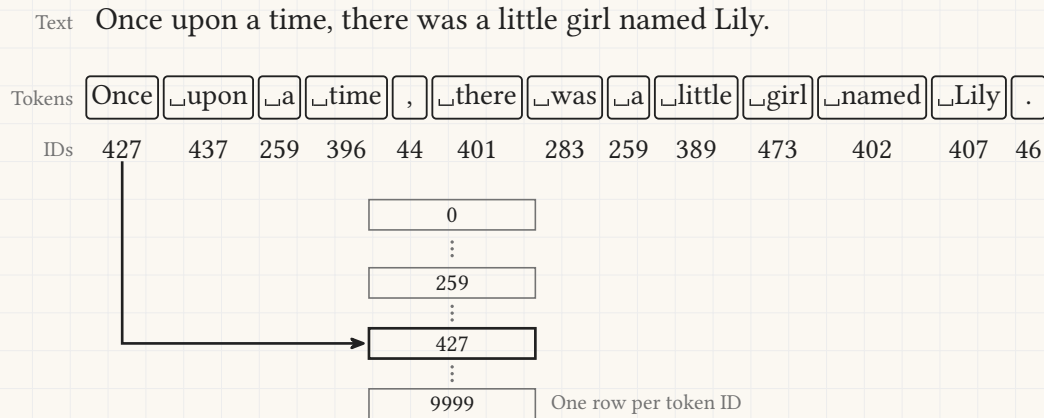
Text Once upon a time, there was a little girl named Lily.

Tokens	Once	␣upon	␣a	␣time	,	␣there	␣was	␣a	␣little	␣girl	␣named	␣Lily	.
IDs	427	437	259	396	44	401	283	259	389	473	402	407	46

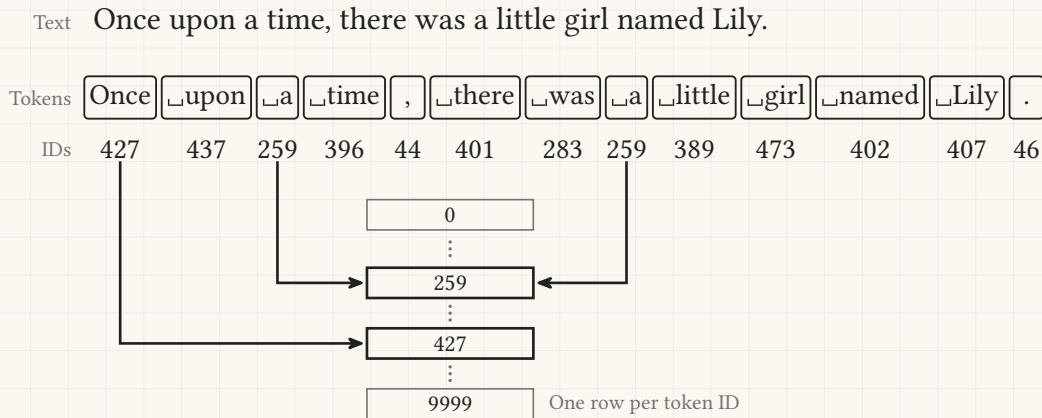


One row per token ID

A model reads token IDs



A model reads token IDs



One token per word

Corpus

Tokens 6

she

sells

seashells

by

the

sea

One token per word

Corpus

Tokens 6

she

sells

seashells

by

the

sea

Vocabulary

she

sells

seashells

by

the

sea

⋮

Every word ever seen:
it keeps growing

One token per word

Corpus

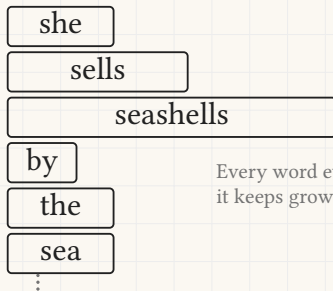
Tokens 6



New text

seashell → UNK No token for it

Vocabulary



Every word ever seen:
it keeps growing

One token per word

Corpus

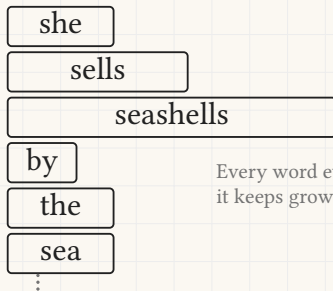


Tokens 6

New text



Vocabulary



Every word ever seen:
it keeps growing

One token per letter

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

One token per letter

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Vocabulary

a b e h l s t y 8 letters

Subwords sit in between

							Tokens	Vocabulary
Words	she	sells	seashells	by	the	sea	6	Unbounded

Subwords sit in between

		Tokens	Vocabulary
Words	she sells seashells by the sea	6	Unbounded
Letters	s h e s e l l s s e a s h e l l s b y t h e s e a	25	8

Subwords sit in between

		Tokens	Vocabulary
Words	she sells seashells by the sea	6	Unbounded
Letters	s h e s e l l s s e a s h e l l s b y t h e s e a	25	8
BPE	she se lls sea she lls b y t he sea	11	14

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Section 2

Byte-pair encoding, by hand

Every letter is a token

Corpus, spaces dropped

s h e s e l l s s e a s h e l l s b y t h e s e a

Every letter is a token

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Every letter is a token

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Vocabulary

a b e h l s t y

Count every adjacent pair

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

Vocabulary

a b e h l s t y

Count every adjacent pair

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

h + e	3	l + s	2
s + e	3	s + h	2
e + a	2	a + s	1
e + l	2	b + y	1
l + l	2	t + h	1

Vocabulary

a b e h l s t y

Count every adjacent pair

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

h + e	3
s + e	3

e + a 2

e + l 2

l + l 2

l + s 2

s + h 2

a + s 1

b + y 1

t + h 1

Vocabulary

a b e h l s t y

Count every adjacent pair

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

h + e	3
s + e	3

e + a 2

e + l 2

l + l 2

l + s 2

s + h 2

a + s 1

b + y 1

t + h 1

Vocabulary

a b e h l s t y

Tie: the greater pair wins, first token, then second

Count every adjacent pair

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

h + e	3
s + e	3

e + a 2

e + l 2

l + l 2

l + s 2

s + h 2

a + s 1

b + y 1

t + h 1

Vocabulary

a b e h l s t y

1 Which pair?

Tie: the greater pair wins, first token, then second

Merge the most frequent pair, everywhere at once

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

h + e	3
s + e	3

e + a 2

e + l 2

l + l 2

l + s 2

s + h 2

a + s 1

b + y 1

t + h 1

Vocabulary

a b e h l s t y

Tie: the greater pair wins, first token, then second

Merge the most frequent pair, everywhere at once

Corpus, spaces dropped

Tokens 25

s h e s e l l s s e a s h e l l s b y t h e s e a

Pair counts

h + e	3
s + e	3

e + a 2

e + l 2

l + l 2

l + s 2

s + h 2

a + s 1

b + y 1

t + h 1

Vocabulary

a b e h l s t y

Tie: the greater pair wins, first token, then second

Merge the most frequent pair, everywhere at once

Corpus, spaces dropped

Tokens 22

s h e se l l s se a s h e l l s b y t h e se a

Pair counts

Vocabulary

a b e h l s t y

1 s + e = se

Recount, then merge again

Corpus, spaces dropped

s h e se l l s se a s h e l l s b y t h e se a

Tokens 22

Pair counts

Vocabulary

a b e h l s t y

1 s + e = se

Recount, then merge again

Corpus, spaces dropped

Tokens 22

s h e se l l s se a s h e l l s b y t h e se a

Pair counts

h + e	3	a + s	1
l + l	2	b + y	1
l + s	2	e + l	1
s + h	2	se + l	1
se + a	2	t + h	1

Vocabulary

a b e h l s t y
1 s + e = se

Recount, then merge again

Corpus, spaces dropped

Tokens 22

s h e se l l s se a s h e l l s b y t h e se a

Pair counts

h + e 3

l + l 2

l + s 2

s + h 2

se + a 2

a + s 1

b + y 1

e + l 1

se + l 1

t + h 1

Vocabulary

a b e h l s t y

1 s + e = se

Recount, then merge again

Corpus, spaces dropped

Tokens **19**

s he se l l s se a s he l l s b y t he se a

Pair counts

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

Four more merges

Corpus, spaces dropped

Tokens 19

s he se l l s se a s he l l s b y t he se a

Pair counts

l + l 2

l + s 2

s + he 2

se + a 2

a + s 1

b + y 1

he + l 1

se + l 1

t + he 1

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 Which pair?

Tie: the greater pair wins, first token, then second

Four more merges

Corpus, spaces dropped

Tokens 19

s he se l l s se a s he l l s b y t he se a

Pair counts

l + l	2
l + s	2
s + he	2
se + a	2
a + s	1

b + y 1

he + l 1

se + l 1

t + he 1

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

Tie: the greater pair wins, first token, then second

Four more merges

Corpus, spaces dropped

Tokens **17**

s he se l l s sea s he l l s b y t he sea

Pair counts

l + l	2
l + s	2
s + he	2

b + y 1

he + l 1

se + l 1

sea + s 1

t + he 1

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

Tie: the greater pair wins, first token, then second

Four more merges

Corpus, spaces dropped

Tokens **15**

she se l l s sea she l l s b y t he sea

Pair counts

l + l 2
l + s 2

b + y 1

se + l 1

sea + she 1

she + l 1

t + he 1

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

Tie: the greater pair wins, first token, then second

Four more merges

Corpus, spaces dropped

Tokens **13**

she se l ls sea she l ls b y t he sea

Pair counts

l + ls 2

t + he 1

b + y 1

se + l 1

sea + she 1

she + l 1

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

Four more merges

Corpus, spaces dropped

Tokens **11**

she se **lls** sea she **lls** b y t he sea

Pair counts

b + y 1
se + lls 1
sea + she 1
she + lls 1
t + he 1

Vocabulary

a b e h l s t y
1 s + e = se
2 h + e = he
3 se + a = sea
4 s + he = she
5 l + s = ls
6 l + ls = **lls**

Stop when no pair occurs twice

Corpus, spaces dropped

Tokens **11**

she se lls sea she lls b y t he sea

Pair counts

b + y 1
se + lls 1
sea + she 1
she + lls 1
t + he 1

Vocabulary

a b e h l s t y
1 s + e = se
2 h + e = he
3 se + a = sea
4 s + he = she
5 l + s = ls
6 l + ls = lls

Stop when no pair occurs twice

Corpus, spaces dropped

Tokens **11**

she se lls sea she lls b y t he sea

Pair counts

b + y	1
se + lls	1
sea + she	1
she + lls	1
t + he	1

Vocabulary

a	b	e	h	l	s	t	y
1	s	+	e	=	se		
2	h	+	e	=	he		
3	se	+	a	=	sea		
4	s	+	he	=	she		
5	l	+	s	=	ls		
6	l	+	ls	=	lls		

Stop when no pair occurs twice

Corpus, spaces dropped

Tokens 11

she se lls sea she lls b y t he sea

Pair counts

b + y	1
se + lls	1
sea + she	1
she + lls	1
t + he	1

Vocabulary

a	b	e	h	l	s	t	y
1	s	+	e	=	se		
2	h	+	e	=	he		
3	se	+	a	=	sea		
4	s	+	he	=	she		
5	l	+	s	=	ls		
6	l	+	ls	=	lls		

} 8 + 6 = 14

Stop when no pair occurs twice

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

Pair counts

b + y	1
se + lls	1
sea + she	1
she + lls	1
t + he	1

Vocabulary

a	b	e	h	l	s	t	y
1	s	+	e	=	se		
2	h	+	e	=	he		
3	se	+	a	=	sea		
4	s	+	he	=	she		
5	l	+	s	=	ls		
6	l	+	ls	=	lls		

8 + 6 = 14

The exercise stops at
a vocabulary of 10,000

The merge list is the tokeniser

Corpus, spaces dropped

Tokens **11**

she se lls sea she lls b y t he sea

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

The merge list is the tokeniser

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

ID to token, 14 entries

0 a

1 b

2 e

3 h

4 l

5 s

6 t

7 y

8 se

9 he

10 sea

11 she

12 ls

13 lls

The merge list is the tokeniser

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

A word not in the corpus

s h e l l s

Vocabulary

a b e h l s t y

ID to token, 14 entries

1 s + e = se
2 h + e = he
3 se + a = sea
4 s + he = she
5 l + s = ls
6 l + ls = lls

0 a
1 b
2 e
3 h
4 l
5 s
6 t
7 y
8 se
9 he
10 sea
11 she
12 ls
13 lls

The merge list is the tokeniser

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

A word not in the corpus

s h e l l s
s he l l s

Merge 2, ID 9

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

ID to token, 14 entries

0	a	
1	b	
2	e	8 se
3	h	9 he
4	l	10 sea
5	s	11 she
6	t	12 ls
7	y	13 lls

The merge list is the tokeniser

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

A word not in the corpus

s h e l l s

s he l l s Merge 2, ID 9

she l l s Merge 4, ID 11

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

ID to token, 14 entries

0	a	
1	b	
2	e	8 se
3	h	9 he
4	l	10 sea
5	s	
6	t	11 she
7	y	12 ls
		13 lls

The merge list is the tokeniser

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

A word not in the corpus

s h e l l s

s he l l s Merge 2, ID 9

she l l s Merge 4, ID 11

she l ls Merge 5, ID 12

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

ID to token, 14 entries

0	a	
1	b	
2	e	8 se
3	h	9 he
4	l	10 sea
5	s	
6	t	11 she
7	y	12 ls
		13 lls

The merge list is the tokeniser

Corpus, spaces dropped

she se lls sea she lls b y t he sea

Tokens 11

A word not in the corpus

s h e l l s

s he l l s Merge 2, ID 9

she l l s Merge 4, ID 11

she l ls Merge 5, ID 12

she lls Merge 6, ID 13

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

ID to token, 14 entries

0	a	
1	b	
2	e	8 se
3	h	9 he
4	l	10 sea
5	s	
6	t	11 she
7	y	12 ls
		13 lls

The merge list is the tokeniser

Corpus, spaces dropped

Tokens **11**

she se lls sea she lls b y t he sea

A word not in the corpus

s h e l l s

s he l l s Merge 2, ID 9

she l l s Merge 4, ID 11

she l ls Merge 5, ID 12

she lls Merge 6, ID 13

she + lls = shells Decoding joins the pieces

Vocabulary

a b e h l s t y

1 s + e = se

2 h + e = he

3 se + a = sea

4 s + he = she

5 l + s = ls

6 l + ls = lls

ID to token, 14 entries

0	a	
1	b	
2	e	8 se
3	h	9 he
4	l	10 sea
5	s	
6	t	11 she
7	y	12 ls
		13 lls

A letter the toy vocabulary is missing

The toy's vocabulary

a b e h l s t y

A letter the toy vocabulary is missing

The toy's vocabulary

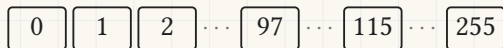
a b e h l s t y

A new word

h e l l o No token for o

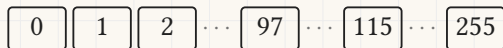
Every character is a sequence of bytes

The 256 byte values



Every character is a sequence of bytes

The 256 byte values



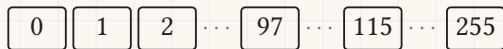
a

s

Letters are bytes already

Every character is a sequence of bytes

The 256 byte values

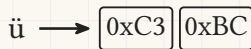


a

s

Letters are bytes already

One letter, two bytes



Byte-level BPE starts from 256 bytes

A new word

h	e	l	l	o
---	---	---	---	---

 The letter o is a byte too

Byte-level BPE starts from 256 bytes

A new word

h e l l o The letter o is a byte too

Base vocabulary

256

Byte-level BPE, as in GPT-2

Over 130,000

BPE over Unicode code points

3

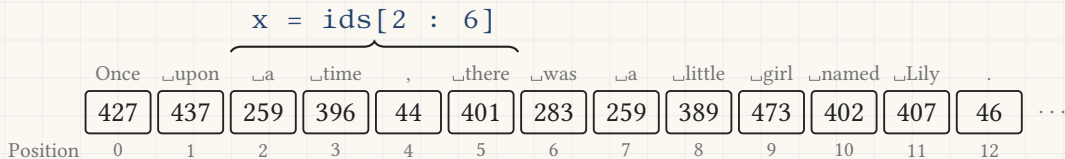
Section 3

From a text file to training examples

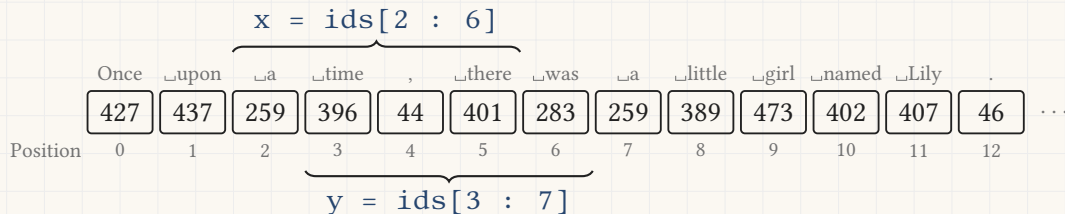
A training example: a window, shifted by one

	Once	upon	a	time	,	there	was	a	little	girl	named	Lily	.	...
	427	437	259	396	44	401	283	259	389	473	402	407	46	
Position	0	1	2	3	4	5	6	7	8	9	10	11	12	

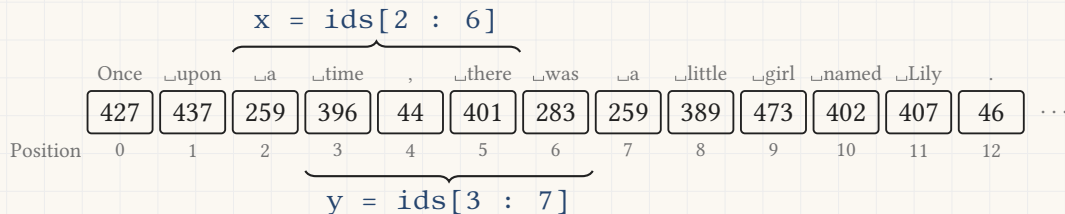
A training example: a window, shifted by one



A training example: a window, shifted by one

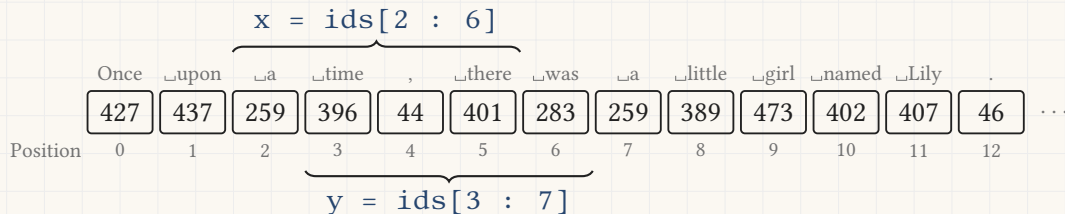


A training example: a window, shifted by one



`x = ids[i : i+T]`

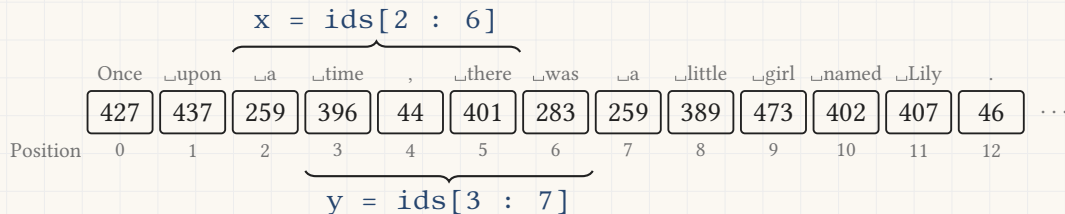
A training example: a window, shifted by one



`x = ids[i : i+T]`

`y = ids[i+1 : i+T+1]`

A training example: a window, shifted by one



$x = \text{ids}[i : i+T]$

$y = \text{ids}[i+1 : i+T+1]$

A batch: B windows,
 $B \times T$ IDs